

Rochester (Zhijie) Lin

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY (Expected) Dec. 2025
Master of Engineering in *Operations Research & Information Engineering*
Concentration: Data Analytics

University of Waterloo, Faculty of Mathematics, Canada Sept. 2019 – Jun. 2024
Bachelor of Mathematics
Double Majors: *Honors Mathematical Finance & Honors Statistics* | Minor: *Pure Mathematics*
Awards: President's Scholarship (2019); Dean's Honors List (2020); Term Distinction (2021–2023)

RELEVANT COURSES

Introduction to Game Theory • Mathematics of Financial Markets • Computational Statistics and Data Analysis • Monte Carlo Simulation • Generalized Linear Models • Mathematical Statistics • Applied Time Series Analysis

WORK EXPERIENCE

Shanghai Huicheng Asset Management Co., Ltd, China Jan. 2024 – Aug. 2024

Trading Desk Support – Government Bond Trading

- Conducted correlation analysis among different products in bond market and identified suitable derivatives to be included in the portfolio to capture instantaneous arbitrage opportunities with Tinysoft.
- Created portfolios consisting of bond ETFs and futures using regression modeling and duration matching, optimizing with mean-variance analysis and risk parity approach to boost the annualized yield by 1.21%.
- Developed portfolios using both delta-neutral and volatility-neutral approaches, analyzed the characteristics of arbitrage opportunities from backtesting results, communicating findings to a team of six traders.

Trading Desk Support – ETF Trading

- Developed and implemented arbitrage strategies (delta-neutral, basis arbitrage, and long-short) and trend following strategies on derivatives to profit from the market fluctuations.
- Utilized Tinysoft and Python to extract Level II data and develop methods for detecting and replacing outliers in option prices affected by trading curbs. Presented trend charts of historical performance for key indicators and selected portfolios during weekly team reviews to support adjustments in future trading plan.
- Tested and implemented a newly automated trading system developed by in-house IT team, calibrating parameters with Webstock—an advanced signal system—to reduce transaction costs by about 12%.
- Visualized 9 custom indicators in real-time trend charts to aid portfolio managers' trading decisions.
- Achieved returns of up to 1,000% through twice a month zero-days-to-expiration (ODTE) option trading opportunities from January to August 2024.

Shanghai Huicheng Asset Management Co., Ltd, China Aug. 2020 – Aug. 2021

Assistant Trader

- Monitored balancing cash, estimated cash substitution, and predicted ETF constituent stock dividend points to optimize the creation and redemption process, resulting in a 32.88% improvement in fund efficiency.
- Utilized proprietary signals and trading system to redeem or create ETFs based on anticipated market trends, leveraging the Limit Up-Limit Down rule and T+2 clearance to maximize profits, achieving an average return of 17.4% from January to August 2021.

PROJECT EXPERIENCE

Analysis of Extreme Events for JP Morgan Chase & Co. Jan. 2023 – Apr. 2023

Quantitative Enterprise Risk Management, University of Waterloo

- Conducted correlation analysis between JP Morgan stock and Gold prices with regression and tail dependence coefficients. Modeled extreme events of JP Morgan with Block Maxima Method and Peaks-over-Threshold.
- Findings:** Applying the Generalized Pareto Distribution Method, the estimations of two risk measures (Value-at-Risk & expected shortfall) yielded the lowest biases to the historical data.

SKILLS

- Computer:** Python (Numpy, Pandas, Matplotlib), R, MATLAB, SQL, LaTeX, Microsoft Excel & PowerPoint
- Financial:** TradingView, Webstock (Trading), Tinysoft (Data Analysis)